General Mathematics — Exercise Sheet 38

Double integrals and the Gaussian

Variant: Statements

Exercise 1 (A curved region requiring two horizontal pieces). Let $D = \{(x, y) : 0 \le x \le 1,\ x^2 \le y \le 2 - x\}$. Describe D in the opposite integration order, splitting it where necessary. Find its area and evaluate $\iint_D (x + 2y)\,\mathrm{d}A$ in both orders. Use Lecture 38's compact Fubini rule and Lecture 26's proper integration methods.

Exercise 2 (Nonlinear coordinates in the positive quadrant). Let D be the positive-quadrant region satisfying $1 \le xy \le 4$ and $1 \le y/x \le 9$. Use $u = xy$ and $v = y/x$ to find its area and $\iint_D (y/x)\,\mathrm{d}A$. Construct the inverse map and check every change-of-variables hypothesis. Use Lectures 8 and 36's differentiation and Jacobian rules, Lecture 38's standing fact M and Lecture 26's elementary integrals.

Exercise 3 (Compact elliptical Gaussian cutoffs). For $a > 0$, let $D_a = \{(x, y) : x^2 + xy + y^2 \le a\}$ and define $I(a) = \iint_{D_a} e^{-(x^2+xy+y^2)}\,\mathrm{d}A$. Find $I(a)$, the limit $L = \lim_{a\to\infty} I(a)$ and the exact cutoff error $L - I(a)$. Use a linear change followed by Lecture 38's compact polar Gaussian value. Use Lecture 36's Jacobian composition and Lecture 22's determinant rule. Every multiple integral in the calculation must have a compact domain.

Exercise 4 (The centroid of a quarter-disc with radial density). On $D = \{(x, y) : x, y \ge 0,\ x^2 + y^2 \le R^2\}$, where $R > 0$, let $\rho(x, y) = \kappa(x^2 + y^2)^{p/2}$, with $\kappa > 0$ and $p \ge 0$. At the origin set the density to κ if $p = 0$ and to zero if $p > 0$. Find its mass and centroid using Lecture 38's definitions and polar rule. Include $p = 0$, and find the centroid limit as $p \to \infty$. Use Lecture 26's power and trigonometric primitives.

Exercise 5 (Differentiation under a proper integral). For $a > -1$, define

$$F(a) = \int_0^{\pi/2} \ln(1 + a\sin^2 t)\,\mathrm{d}t.$$

Justify the parameter-differentiation clause of standing fact M near each allowed a, compute $F'(a)$ including $a = 0$, and find $F(a)$. Use the auxiliary integral $J(a) = \int_0^{\pi/2} (1 + a\sin^2 t)^{-1}\,\mathrm{d}t$. Use Lecture 26's substitution, Lecture 27's ordinary improper limits, Lecture 8's arctangent derivative and Lecture 9's constant-difference theorem.

Exercise 6 (Compact cutoffs and incompatible limiting orders ⋆). For $x, y > 0$, let $f(x, y) = (x^2 - y^2)/(x^2 + y^2)^2$. For $0 < \varepsilon, \delta < 1$, define the proper compact integral

$$K(\varepsilon, \delta) = \int_{\delta}^{1} \int_{\varepsilon}^{1} f(x, y)\, \mathrm{d}x\, \mathrm{d}y.$$

Find K explicitly. Compute its limit when $\varepsilon \to 0^+$ first and $\delta \to 0^+$ second, and then in the opposite order. Also compute the joint approaches $\delta = \varepsilon$ and $\delta = \varepsilon^2$. Explain why the results do not contradict M and do not define one joint cutoff limit. Use Lectures 8 and 26's primitives and Lecture 27's ordinary endpoint-limit conventions.